

ADVANCED

Stage 6

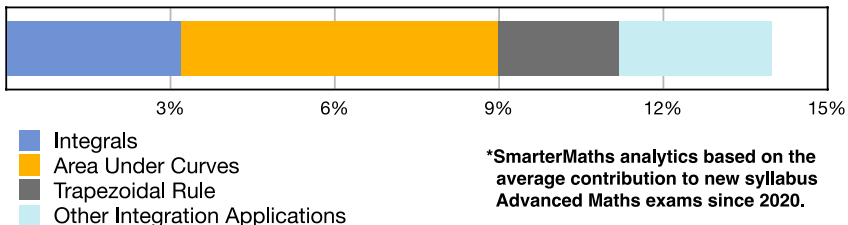
Calculus (Adv), C4 Integration (Adv)

Areas Under Curves Part 2 (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 66 minutes (based on allocation of 1.5 minutes per mark)

C4 Integration



HISTORICAL CONTRIBUTION

- C4 Integration* has contributed a substantial 14.0% to new syllabus Advanced exams since it was introduced in 2020.
- This topic has been split into four sub-topics for analysis purposes: *1-Integrals (3.2%)*, *2-Areas Under Curves (5.8%)*, *3-Trapezoidal Rule (2.2%)* and *4-Other Integration Examples (2.8%)*.
- This analysis looks at *Areas Under Curves*.

HSC ANALYSIS - What to expect and common pitfalls

- Areas Under Curves (5.8%)* has attracted an average of two questions per new syllabus exam, appearing in both multiple choice and longer answer sections.
- This topic area can be examined using a very broad range of underlying equations, including quadratic, cubic, exponential, hyperbola and trig functions.
- The most common question type requires students to find areas between given curves and straight lines (although the area between 2 quadratics was asked in 2024 and 2020).
- Important revision questions: *2021 Adv 28* combined Area Under Curves with transformations and *2022 Adv 28* looked at circle functions in this context → both caused problems.
- Definite integrals defining areas for functions that cannot be integrated has been asked via multiple choice 3 times in new syllabus exams, all producing sub-50% mean marks.

Questions

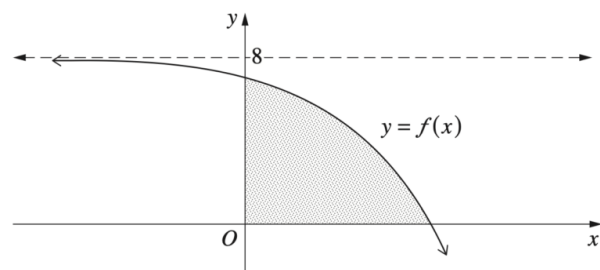
1. Calculus, 2ADV C4 2016 HSC 9 MC

What is the value of $\int_{-3}^2 |x + 1| \, dx$?

- $\frac{5}{2}$
- $\frac{11}{2}$
- $\frac{13}{2}$
- $\frac{17}{2}$

2. Calculus, 2ADV C4 2021 HSC 28

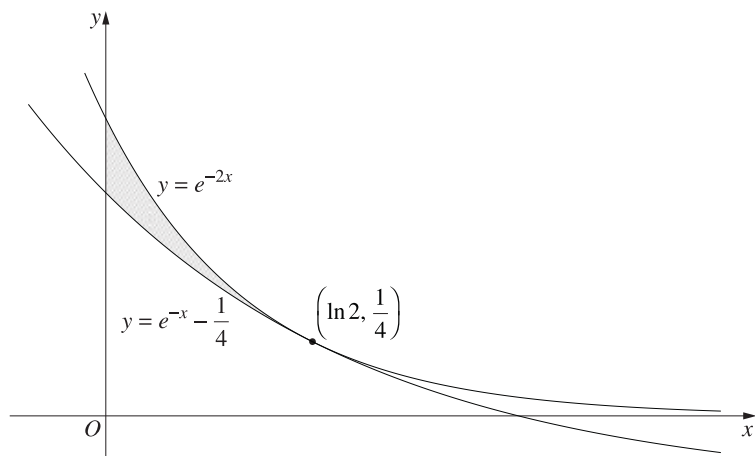
The region bounded by the graph of the function $f(x) = 8 - 2^x$ and the coordinate axes is shown



- Show that the exact area of the shaded region is given by $24 - \frac{7}{\ln 2}$. **(3 marks)**
- A new function $g(x)$ is found by taking the graph of $y = -f(-x)$ and translating it by 5 units to the right.
Sketch the graph of $y = g(x)$ showing the x -intercept and the asymptote. **(2 marks)**
- Hence, find the exact value of $\int_2^5 g(x) \, dx$. **(1 mark)**

3. Calculus, 2ADV C4 2023 HSC 32

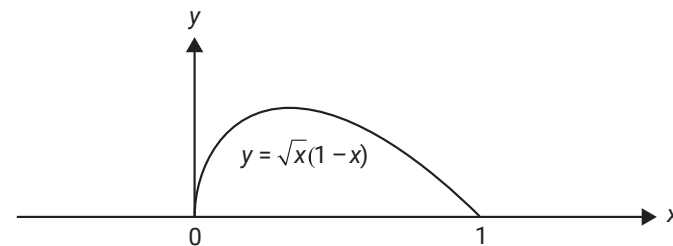
The curves $y = e^{-2x}$ and $y = e^{-x} - \frac{1}{4}$ intersect at exactly one point as shown in the diagram. The point of intersection has coordinates $\left(\ln 2, \frac{1}{4}\right)$. (Do NOT prove this.)



- Show that the area bounded by the two curves and the y -axis, as shaded in the diagram, is $\frac{1}{4}\ln 2 - \frac{1}{8}$. (3 marks)
- Find the values of k such that the curves $y = e^{-2x}$ and $y = e^{-x} + k$ intersect at two points. (3 marks)

4. Calculus, 2ADV C3 SM-Bank 7

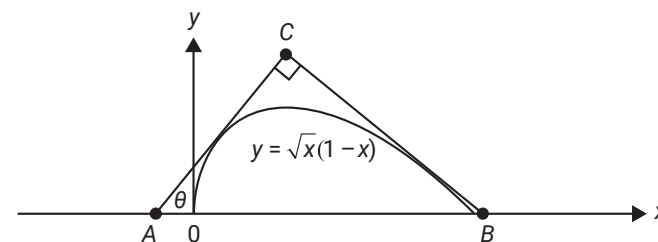
The graph of $f(x) = \sqrt{x}(1-x)$ for $0 \leq x \leq 1$ is shown below.



- Calculate the area between the graph of $f(x)$ and the x -axis. (2 marks)
- For x in the interval $(0, 1)$, show that the gradient of the tangent to the graph of $f(x)$ is $\frac{1-3x}{2\sqrt{x}}$. (1 mark)

The edges of the right-angled triangle ABC are the line segments AC and BC , which are tangent to the graph of $f(x)$, and the line segment AB , which is part of the horizontal axis, as shown below.

Let θ be the angle that AC makes with the positive direction of the horizontal axis.



- Find the equation of the line through B and C in the form $y = mx + c$, for $\theta = 45^\circ$. (3 marks)

5. Calculus, 2ADV C3 2009 HSC 10

Let $f(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$

a. Show that the graph of $y = f(x)$ has no turning points. (2 marks)

b. Find the point of inflection of $y = f(x)$. (1 mark)

c. i. Show that $1 - x + x^2 - \frac{1}{1+x} = \frac{x^3}{1+x}$ for $x \neq -1$. (1 mark)

ii. Let $g(x) = \ln(1+x)$.

Use the result in part c.i. to show that $f'(x) \geq g'(x)$ for all $x \geq 0$. (2 marks)

d. Sketch the graphs of $y = f(x)$ and $y = g(x)$ for $x \geq 0$. (2 marks)

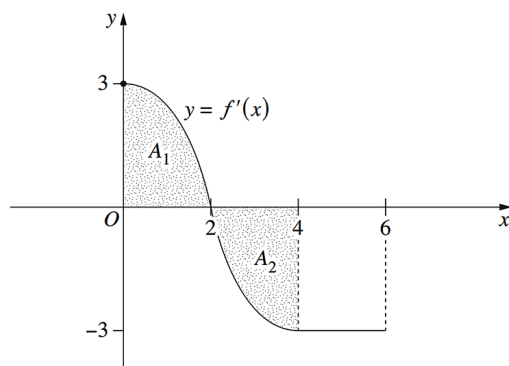
e. Show that $\frac{d}{dx}[(1+x)\ln(1+x) - (1+x)] = \ln(1+x)$. (2 marks)

f. Find the area enclosed by the graphs of $y = f(x)$ and $y = g(x)$, and the straight line $x = 1$. (2 marks)

6. Calculus, 2ADV C3 2010 HSC 9b

Let $y = f(x)$ be a function defined for $0 \leq x \leq 6$, with $f(0) = 0$.

The diagram shows the graph of the derivative of f , $y = f'(x)$.



The shaded region A_1 has area 4 square units. The shaded region A_2 has area 4 square units.

i. For which values of x is $f(x)$ increasing? (1 mark)

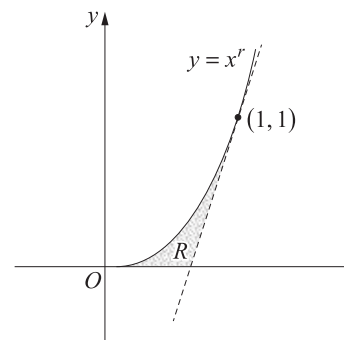
ii. What is the maximum value of $f(x)$? (1 mark)

iii. Find the value of $f(6)$. (1 mark)

iv. Draw a graph of $y = f(x)$ for $0 \leq x \leq 6$. (2 marks)

7. Calculus, 2ADV C4 2019 HSC 16c

The diagram shows the region R , bounded by the curve $y = x^r$, where $r \geq 1$, the x -axis and the tangent to the curve at the point $(1, 1)$.



i. Show that the tangent to the curve at $(1, 1)$ meets the x -axis at

$$\left(\frac{r-1}{r}, 0\right). \quad (2 \text{ marks})$$

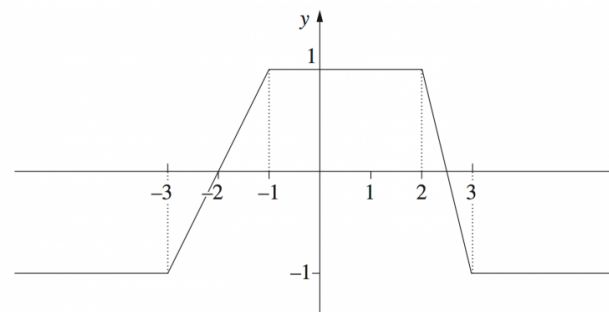
ii. Using the result of part (i), or otherwise, show that the area of the region R is

$$\frac{r-1}{2r(r+1)}. \quad (2 \text{ marks})$$

iii. Find the exact value of r for which the area of R is a maximum. (3 marks)

8. Calculus, 2ADV C4 2013 HSC 14d

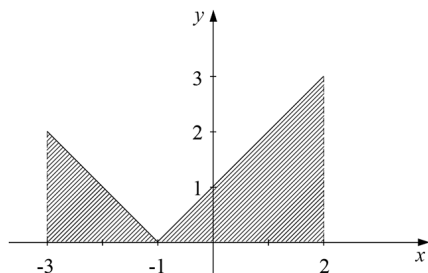
The diagram shows the graph $f(x)$.



What is the value of a , where $a > 0$, so that $\int_{-a}^a f(x) dx = 0$? (1 mark)

Worked Solutions

1. Calculus, 2ADV C4 2016 HSC 9 MC



◆◆◆ Mean mark 19%.

$$\begin{aligned}\int_{-3}^2 |x + 1| \, dx &= \text{Area of 2 triangles} \\ &= \frac{1}{2} \times 2 \times 2 + \frac{1}{2} \times 3 \times 3 \\ &= \frac{13}{2} \\ &\Rightarrow C\end{aligned}$$

Worked Solutions

2. Calculus, 2ADV C4 2021 HSC 28

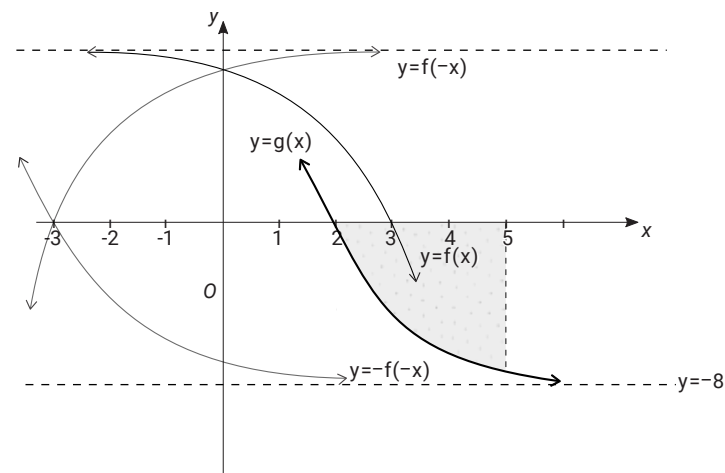
a. x -intercept occurs when

$$8 - 2^x = 0 \Rightarrow x = 3$$

$$\begin{aligned}\text{Area} &= \int_0^3 8 - 2^x \, dx \\ &= \left[8x - \frac{2^x}{\ln 2} \right]_0^3 \\ &= 24 - \frac{8}{\ln 2} - \left(0 - \frac{1}{\ln 2} \right) \\ &= 24 - \frac{8}{\ln 2} + \frac{1}{\ln 2} \\ &= 24 - \frac{7}{\ln 2} \quad \text{u}^2\end{aligned}$$

b.

◆ Mean mark part (b) 48%.



$y = f(-x) \rightarrow$ reflect $y = f(x)$ in the y -axis

$y = -f(-x) \rightarrow$ reflect $y = f(-x)$ in the x -axis

c. $\int_2^5 g(x) \, dx$ is the same area as found in part (a) except it is below the x -axis.

◆◆◆ Mean mark part (c) 13%.

$$\therefore \int_2^5 g(x) \, dx = \frac{7}{\ln 2} - 24$$

3. Calculus, 2ADV C4 2023 HSC 32

$$\begin{aligned}
 \text{a. } A &= \int_0^{\ln 2} e^{-2x} - (e^{-x} - \frac{1}{4}) dx \\
 &= \left[-\frac{1}{2}e^{-2x} + e^{-x} + \frac{1}{4}x \right]_0^{\ln 2} \\
 &= \left[\left(-\frac{1}{2}e^{-2\ln 2} + e^{-\ln 2} + \frac{1}{4}\ln 2 \right) - \left(-\frac{1}{2}e^0 + e^0 - 0 \right) \right] \\
 &= \left[\left(-\frac{1}{2}e^{\ln(2^{-2})} + e^{\ln(2^{-1})} + \frac{1}{4}\ln 2 + \frac{1}{2} - 1 \right) \right] \\
 &= \left[\left(-\frac{1}{2}e^{\ln \frac{1}{4}} + e^{\ln \frac{1}{2}} + \frac{1}{4}\ln 2 - \frac{1}{2} \right) \right] \\
 &= -\frac{1}{2} \times \frac{1}{4} + \frac{1}{2} + \frac{1}{4}\ln 2 - \frac{1}{2} \\
 &= \frac{1}{4}\ln 2 - \frac{1}{8}
 \end{aligned}$$

b. Intersection occurs when

◆◆ Mean mark (b) 14%.

$$\begin{aligned}
 e^{-2x} &= e^{-x} + k \\
 e^{-2x} - e^{-x} - k &= 0
 \end{aligned}$$

$$\text{Let } X = e^{-x}$$

$$X^2 - X - k = 0$$

$$\begin{aligned}
 X &= \frac{1 \pm \sqrt{1^2 - 4(1)(-k)}}{2} \\
 &= \frac{1 \pm \sqrt{1+4k}}{2}
 \end{aligned}$$

$$2 \text{ solutions} \Rightarrow \Delta > 0$$

$$1 + 4k > 0 \Rightarrow k > -\frac{1}{4}$$

$$\text{Since } X = e^{-x} > 0$$

$$\frac{1 - \sqrt{1+4k}}{2} > 0$$

$$1 - \sqrt{1+4k} > 0$$

$$\sqrt{1+4k} < 1$$

$$k < 0$$

$$\therefore -\frac{1}{4} < k < 0$$

4. Calculus, 2ADV C3 SM-Bank 7

$$\begin{aligned}
 \text{i. Area} &= \int_0^1 (\sqrt{x} - x\sqrt{x}) dx \\
 &= \int_0^1 \left(x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx \\
 &= \left[\frac{2}{3}x^{\frac{3}{2}} - \frac{2}{5}x^{\frac{5}{2}} \right]_0^1 \\
 &= \left(\frac{2}{3} - \frac{2}{5} \right) - (0 - 0) \\
 &= \frac{10}{15} - \frac{6}{15} \\
 &= \frac{4}{15} \text{ units}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. } f(x) &= x^{\frac{1}{2}} - x^{\frac{3}{2}} \\
 f'(x) &= \frac{1}{2}x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} \\
 &= \frac{1}{2\sqrt{x}} - \frac{3\sqrt{x}}{2} \\
 &= \frac{1-3x}{2\sqrt{x}} \text{ .. as required.}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii. } m_{AC} &= \tan 45^\circ = 1 \\
 \Rightarrow m_{BC} &= -1 \quad (m_{BC} \perp m_{AC})
 \end{aligned}$$

$$\text{At point of tangency of } BC, \quad f'(x) = -1$$

$$\frac{1-3x}{2\sqrt{x}} = -1$$

$$1 - 3x = -2\sqrt{x}$$

$$3x - 2\sqrt{x} - 1 = 0$$

$$\text{Let } a = \sqrt{x},$$

$$3a^2 - 2a - 1 = 0$$

$$(3a+1)(a-1) = 0$$

$$a = 1 \text{ or } -\frac{1}{3}$$

$$\therefore \sqrt{x} = 1 \quad \text{or} \quad \sqrt{x} = -\frac{1}{3} \quad (\text{no solution})$$

$$x = 1$$

$$f(1) = \sqrt{1}(1-1) = 0 \Rightarrow B(1,0)$$

◆◆◆ Mean mark (Vic) part (iii) 20%.
MARKER'S COMMENT: Most successful answers introduced a pronumeral such as $a = \sqrt{x}$ to solve.

Equation of BC , $m = -1$, through $(1,0)$:

$$y - 0 = -1(x - 1)$$

$$y = -x + 1$$

5. Calculus, 2ADV C3 2009 HSC 10

a. $f(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$

Turning points when $f'(x) = 0$

$$f'(x) = 1 - x + x^2$$

$$x^2 - x + 1 = 0$$

$$\text{Since } \Delta = b^2 - 4ac$$

$$= (-1)^2 - 4 \times 1 \times 1$$

$$= -3 < 0 \Rightarrow \text{No solution}$$

$\therefore f(x)$ has no turning points

b. P.I. when $f''(x) = 0$

$$f''(x) = -1 + 2x = 0$$

$$2x = 1$$

$$x = \frac{1}{2}$$

Check for change in concavity

$$f''\left(\frac{1}{4}\right) = -\frac{1}{2} < 0$$

$$f''\left(\frac{3}{4}\right) = \frac{1}{2} > 0$$

$\Rightarrow$ Change in concavity

$$\therefore \text{P.I. at } x = \frac{1}{2}$$

$$\begin{aligned} f\left(\frac{1}{2}\right) &= \frac{1}{2} - \frac{\left(\frac{1}{2}\right)^2}{2} + \frac{\left(\frac{1}{2}\right)^3}{3} \\ &= \frac{1}{2} - \frac{1}{8} + \frac{1}{24} \\ &= \frac{5}{12} \end{aligned}$$

$\therefore$ Point of Inflection at $\left(\frac{1}{2}, \frac{5}{12}\right)$

c.i. Show $1 - x + x^2 - \frac{1}{1+x} = \frac{x^3}{1+x}, \quad x \neq -1$

$$\begin{aligned} \text{LHS} &= \frac{1+x}{1+x} - \frac{x(1+x)}{1+x} + \frac{x^2(1+x)}{(1+x)} - \frac{1}{1+x} \\ &= \frac{1+x - x - x^2 + x^2 + x^3 - 1}{1+x} \end{aligned}$$

♦♦ Mean mark 28% for all of Q10
(note that data for each question part
is not available).

$$= \frac{x^3}{1+x} \quad \dots \text{ as required}$$

c.ii. Let $g(x) = \ln(1+x)$

$$g'(x) = \frac{1}{1+x}$$

$$f'(x) - g'(x) = 1 - x + x^2 - \frac{1}{1+x}$$

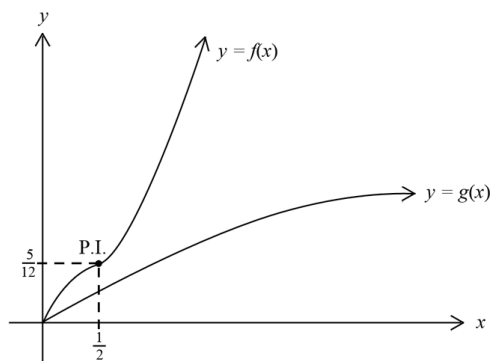
$$= \frac{x^3}{1+x} \quad (\text{using part (i)})$$

Since $\frac{x^3}{1+x} \geq 0$ for $x \geq 0$

$$f'(x) - g'(x) \geq 0$$

$$f'(x) \geq g'(x) \text{ for } x \geq 0$$

d.



MARKER'S COMMENT: When 2 graphs are drawn on the same set of axes, you **must** label them.

e. Show $\frac{d}{dx}[(1+x)\ln(1+x) - (1+x)] = \ln(1+x)$

Using $\frac{d}{dx}uv = uv' + vu'$

$$\text{LHS} = (1+x) \times \frac{1}{1+x} + \ln(1+x) \times 1 + -1$$

$$= 1 + \ln(1+x) - 1$$

$$= \ln(1+x)$$

$$= \text{RHS} \quad \dots \text{ as required}$$

f. Area = $\int_0^1 f(x) - g(x) \, dx$

$$= \int_0^1 \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \ln(x+1) \right) dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{12} - (1+x)\ln(1+x) + (1+x) \right]_0^1$$

(using part (e) above)

$$= \left[\left(\frac{1}{2} - \frac{1}{6} + \frac{1}{12} - (2)\ln 2 + 2 \right) - (\ln 1 + 1) \right]$$

$$= \frac{5}{12} - 2\ln 2 + 2 - 1$$

$$= 1 - \frac{5}{12} - 2\ln 2 \quad \text{u}^2$$

6. Calculus, 2ADV C3 2010 HSC 9b

- i. $f(x)$ is increasing when $f'(x) > 0$

From the graph

$f(x)$ is increasing when $0 \leq x < 2$

- ii. $f'(x) = 0$ when $x = 2$

$\therefore$ MAX at $x = 2$

$$\int_0^2 f'(x) \, dx = 4 \quad (\text{given since } A_1 = 4)$$

We also know

$$\begin{aligned} \int_0^2 f'(x) \, dx &= [f(x)]_0^2 \\ &= f(2) - f(0) \\ &= f(2) \quad (\text{since } f(0) = 0) \\ \Rightarrow f(2) &= 4 \end{aligned}$$

$\therefore$ MAX value of $f(x) = 4$

$$\begin{aligned} \text{iii. } \int_0^4 f'(x) \, dx &= A_1 - A_2 \\ &= 0 \end{aligned}$$

We also know

$$\begin{aligned} \int_0^4 f'(x) \, dx &= \int_2^4 f'(x) \, dx + \int_0^2 f'(x) \, dx \\ &= [f(x)]_2^4 + 4 \\ &= f(4) - f(2) + 4 \quad (\text{note } f(2) = 4) \\ &= f(4) \\ \Rightarrow f(4) &= 0 \end{aligned}$$

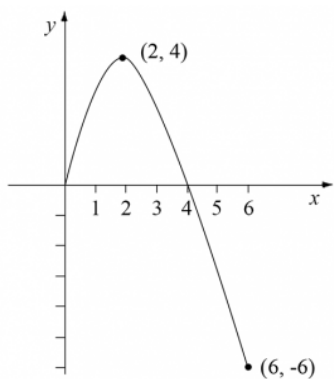
Gradient $= -3$ from $x = 4$ to $x = 6$

$$\begin{aligned} \therefore f(6) &= -3(6 - 4) \\ &= -6 \end{aligned}$$

- iv.

◆◆◆ Parts (ii) and (iii) proved particularly difficult for students with mean marks of 12% and 11% respectively.

◆◆ Mean mark 28%
EXAM TIP: Clearly identify *THE EXTREMES* when given a defined domain. In this case, the origin is obvious graphically, and the other extreme at $x = 6$, is *CLEARLY LABELLED!*



7. Calculus, 2ADV C4 2019 HSC 16c

i. $y = x^r$

$$\frac{dy}{dx} = rx^{r-1}$$

When $x = 1$, $\frac{dy}{dx} = r$

Equation of tangent:

$$y - 1 = r(x - 1)$$

$$y = rx - r + 1$$

When $y = 0$:

$$rx - r + 1 = 0$$

$$rx = r - 1$$

$$x = \frac{r-1}{r}$$

$\therefore$ Tangent meets x -axis at $\left(\frac{r-1}{r}, 0\right)$

♦♦ Mean mark part (i) 31%.

ii. Area under curve

$$= \int_0^1 x^r$$

$$= \left[\frac{1}{r+1} \cdot x^{r+1} \right]_0^1$$

$$= \frac{1}{r+1} \times 1^{r+1} - 0$$

$$= \frac{1}{r+1}$$

♦♦♦ Mean mark part (ii) 21%.

Area under tangent

$$= \frac{1}{2} \times b \times h$$

$$= \frac{1}{2} \left(1 - \frac{r-1}{r} \right) \times 1$$

$$= \frac{1}{2} \left(1 - \frac{r-1}{r} \right)$$

$$\therefore R = \frac{1}{r+1} - \frac{1}{2} \left(1 - \frac{r-1}{r} \right)$$

$$= \frac{1}{r+1} - \frac{1}{2r} [r - (r-1)]$$

$$= \frac{1}{r+1} - \frac{1}{2r}$$

$$= \frac{2r-(r+1)}{2r(r+1)}$$

$$= \frac{r-1}{2r(r+1)}$$

iii. $R = \frac{r-1}{2r(r+1)} = \frac{r-1}{2r^2+2r}$

$$\frac{dR}{dr} = \frac{(2r^2+2r) \times 1 - (r-1)(4r+2)}{(2r^2+2r)^2}$$

$$= \frac{2r^2+2r-4r^2-2r+4r+2}{(2r^2+2r)^2}$$

$$= \frac{-2r^2+4r+2}{(2r^2+2r)^2}$$

$$= \frac{-2(r^2-2r-1)}{(2r^2+2r)^2}$$

Find r when $\frac{dR}{dr} = 0$:

$$r^2-2r-1 = 0$$

$$r = \frac{2 \pm \sqrt{(-2)^2 - 4 \times 1 \times (-1)}}{2}$$

$$= \frac{2 \pm \sqrt{8}}{2}$$

$$= 1 + \sqrt{2} \quad (r \geq 1)$$

r	1	$1 + \sqrt{2}$	3
$\frac{dR}{dr}$	$\frac{1}{4}$	0	$-\frac{1}{144}$

$\therefore R_{\max}$ occurs when $r = 1 + \sqrt{2}$

♦♦ Mean mark part (iii) 23%.

8. Calculus, 2ADV C4 2013 HSC 14d

If $\int_{-a}^a f(x) \, dx = 0$

We know the area below the curve and above the x -axis = area above the curve and below the x -axis.

By inspection, we can see

$$\int_{-3}^{-1} f(x) \, dx = 0 \quad \text{and} \quad \int_2^3 f(x) \, dx = 0$$

We need $\int_3^a f(x) \, dx + \int_{-a}^{-3} f(x) \, dx = -3$

because $\int_{-1}^2 f(x) \, dx = 3$

$\Rightarrow$ Since areas have height = 1, each must be 1.5 units wide.
 $\therefore a = 4.5$

♦♦♦ A devilish 1-mark question mid-paper that had a mean mark of just 12%.

MARKER'S COMMENT: The fact that this question was worth only 1 mark means that it is not necessary for students to show any detailed working.